

AN2DL - First Challenge Report

four_loops

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1 Introduction

This project tackles *multivariate time series classification* for automatic prediction of human pain levels. Each sample is a 160-step sequence of body-joint angles, pain-survey indicators, and static attributes. The task is to assign each sequence to one of three classes: *no pain*, *low pain*, or *high pain*.

We design a model trained from scratch, without pretrained weights, and focus on:

- reshaping long-format data into fixed-size tensors;
- standardizing features consistently across time;
- using a compact **1D CNN + BiLSTM** architecture;
- handling class imbalance through **SMOTE**;
- using **focal loss** and a macro F1 callback.

Hybrid CNN-BiLSTM architectures have shown strong performance on other sequence prediction tasks [1], which motivates our choice.

2 Problem Analysis

The data is provided in long format, one row per time step. After grouping by `sample_index` and

sorting by time, each sequence becomes a tensor of shape $(160 \times d)$, where d collects joint angles, survey features, and static variables broadcast along the temporal axis.

The main challenges are:

- **Class imbalance:** the *high pain* class is rare.
- **Temporal structure:** pain-related patterns are spread over 160 steps.
- **Heterogeneous features:** joints, surveys, and static attributes have different scales and meanings.
- **Label noise:** self-reported pain may blur boundaries, especially between *no* and *low pain*.

We assume that motion patterns encode pain information and that a CNN+BiLSTM hybrid can learn both local and medium-range temporal dependencies. We also expect that explicit class rebalancing (SMOTE, focal loss) is necessary to obtain good macro F1.

3 Method

Preprocessing

We first clean categorical limb-count fields (`n_legs`, `n_hands`, `n_eyes`) by mapping textual values such as

“one+peg_leg” to integers and convert other object columns to numeric, filling missing values with zero.

Static features are aggregated per sample via group-by on `sample_index`. We add binary flags (`missing_leg`, `missing_hand`, `missing_eye`, `missing_any_limb`) and align static feature spaces between train and test.

For each sample we:

- stack joint features over time to get a $(160 \times n_{\text{joints}})$ matrix;
- concatenate static features (broadcast along the time dimension).

This yields $X \in R^{N \times 160 \times d}$. Labels are taken from the train labels file, encoded with `LabelEncoder`, and converted to one-hot vectors.

We standardize features by flattening $(160, d)$ into one vector per sample, fitting a `StandardScaler` on the training split, and reshaping back to $(160, d)$.

Class imbalance handling

On the training split we apply Synthetic Minority Over-sampling Technique (SMOTE) on flattened sequences. This produces a balanced training distribution for the three pain classes while keeping validation data unchanged.

Other imbalance strategies tested include class weights and focal-loss parameter variations. The best model, however, uses *only* SMOTE, hence the name `03_SMOTE_Only`.

Model Architecture

The network is a single-stream CNN+BiLSTM:

- Input: sequence of length 160 with d features.
- 1D convolution (ReLU + BatchNorm).
- Max pooling over time, followed by dropout or spatial dropout.
- Bidirectional LSTM with outputs at all time steps.
- Global average pooling across time.
- Dense layer with ReLU and L_1 – L_2 regularization, BatchNorm, and dropout.

- Final dense layer with softmax over three classes.

This architecture is intentionally small to reduce overfitting while still modeling temporal dynamics.

Optimization

We split the data into training and validation sets with an 80/20 stratified split. On the training part we apply scaling, SMOTE, and (optionally) augmentation; the validation part is only scaled.

The optimizer is AdamW with weight decay and gradient clipping. The loss is categorical focal loss with moderate focusing parameters (γ, α) tuned per experiment. A `ReduceLROnPlateau` callback reduces the learning rate on validation-loss plateau. A custom callback monitors validation macro F1, saves the best weights, and stops training if macro F1 does not improve for several epochs.

4 Experiments and Results

We evaluate 25 configurations. They differ along four main axes:

- imbalance handling: no tricks, SMOTE, class weights, or both;
- augmentation: none vs light/medium/heavy Gaussian noise;
- regularization: standard vs spatial dropout and stronger L_1 – L_2 ;
- architecture and loss: more CNN layers and different focal-loss parameters.

Common hyperparameters are:

- batch size 64,
- up to 500 epochs,
- learning rate 5×10^{-4} and weight decay 10^{-5} ,
- early stopping on validation macro F1.

The primary selection metric is macro F1-score on the validation set; weighted F1-score is reported as a secondary indicator.

Table 1: Top 5 Experiments Sorted by Macro-F1

Experiment	Macro-F1	Weighted-F1
03_SMOTE_Only	0.96810	0.98340
24_Balanced_AllRound	0.96538	0.98473
17_SMOTE_Aug_SpatialDrop	0.96309	0.97755
25_KitchenSink_Everything	0.96013	0.97053
22_SMOTE_2CNN_Moderate	0.95834	0.97791

The simplest imbalance-aware configuration, **03_SMOTE_Only**, is the best in terms of macro F1. More complex settings with additional tricks (augmentation, spatial dropout, stronger focal loss, deeper CNN) provide only marginal gains in weighted F1 and slightly worse macro F1.

The final selected model is **03_SMOTE_Only**. It uses SMOTE on the training set, no class weights, no augmentation, and a single convolutional layer followed by BiLSTM. On the validation split it achieves:

- **macro F1-score: 0.968,**
- **weighted F1-score: 0.983.**

Per-class metrics show high recall and F1 for the minority *high pain* class compared to baselines without SMOTE. Most residual errors are confusions between *no pain* and *low pain*, which is expected given their similar feature profiles.

Submission predictions are generated using the checkpoint with the best validation macro F1 and formatted according to the challenge rules.

5 Discussion

The experiments confirm that class-imbalance handling is the main driver of performance. SMOTE applied on flattened sequences is sufficient to equalise the label distribution and yields most of the improvement over the naive baselines. Focal loss further helps by focusing learning on hard examples, but heavy regularisation and strong augmentation bring only limited benefits.

The chosen CNN+BiLSTM model is relatively small in terms of parameters yet reaches high macro and weighted F1-scores. Convolutions capture local

motion patterns, while the recurrent layer aggregates temporal context. This is consistent with results reported in other domains for CNN-BiLSTM hybrids [1] and suggests that, for this dataset, a carefully tuned medium-sized model is preferable to an aggressively large architecture.

Limitations mainly come from data quality and coverage. Labels are self-reported pain estimates, which can be noisy, and the number of *high pain* samples is still limited even after oversampling. Feature scaling is also uniform across all variables, although joints, surveys, and static attributes may require different treatment. Finally, we did not explore more advanced sequence-level augmentations or domain-specific priors.

6 Conclusions

We presented a full pipeline for pain-level recognition from multivariate time series, from cleaning and tensor construction to training and model selection. The final system combines standardized features, a compact CNN+BiLSTM architecture, SMOTE rebalancing, and focal-loss training monitored by a macro F1 callback.

Among 25 variants, the simplest imbalance-aware configuration, **03_SMOTE_Only**, achieves the best validation macro F1-score, outperforming more complex setups with additional tricks. This indicates that getting the label distribution right is more important than stacking many regularisation components.

Future work includes testing transformer-based and deeper recurrent architectures, exploring sequence-level augmentations, and integrating static metadata with attention mechanisms. Calibration and uncertainty estimation would also be important steps toward reliable use of such models in clinical or real-world monitoring scenarios.

References

- [1] L. Shan, Y. Liu, M. Tang, M. Yang, and X. Bai. Cnn-bilstm hybrid neural networks with attention mechanism for well log prediction. *Journal of Petroleum Science and Engineering*, 205:108838, 2021.